

SAWSAN ABDULBARI

Data Scientist & Machine Learning Engineer | AI Innovation

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Biography

I build full-stack AI and ML systems that ship production machine learning, generative AI, and data software used by real teams to drive measurable outcomes and automation. I have designed and delivered a wide range of AI- and ML-based solutions and systems abroad—working with organizations and companies across healthcare, sustainability, finance/trading tools, education, knowledge systems, and digital consultancy—in Finland, Egypt, and globally distributed remote teams.

That work spans end-to-end delivery: predictive modeling and time series, RAG and generative AI, computer vision, interactive dashboards, and production web apps—from Omdena chapter builds and enterprise RAG challenges to Demos Helsinki's Connected Knowledge System, PIRHA healthcare UX, and product systems such as DataDiwan OS and DataDiwan.

Trilingual professional (Arabic, English, Finnish) with an entrepreneurial background. Before moving into engineering, I spent over a decade as an operator including 7 years as a founder, plus senior management and quality control roles. That experience shapes how I build AI today: focused on real-world impact, adoption, and audit-ready governance rather than accuracy in isolation.

Key differentiators: international delivery across diverse sectors and organizations, production ML and GenAI systems, global collaboration (15+ contributor teams), mentoring and curriculum development, and 7 years of founder/operator experience.

Work experience

Junior Expert specializing in AI

Aug 2026 to present

Demos Helsinki · Helsinki Metropolitan Area · Hybrid

Applying AI expertise to Demos Helsinki: Connected Knowledge System (CKS), contributing to the development and delivery of AI-powered knowledge solutions.

Artificial Intelligence · Machine Learning · Connected Knowledge System

Data Scientist and Machine Learning Engineer

Incoming — anticipated Dec 2026

University of Helsinki · Helsinki, Finland

Incoming role: data science and machine learning engineering at the University of Helsinki.

Data Science · Machine Learning · Python

Data Scientist and Machine learning Engineer

Jun 2026 to Aug 2026

Dashtera · Finland

Built and deployed data science and machine learning solutions, applying end-to-end ML workflows to support Dashtera's products and services.

Data Science · Machine Learning · Python

Machine Learning Engineer

2024 to present

Omdena · Palo Alto, CA (Remote)

Collaborating on AI innovation in globally distributed teams. Researching and implementing RAG architecture for enterprise applications.

Python · RAG Architecture · Machine Learning · AI Research

Student Assistant

2024 to 2025

Häme University of Applied Sciences (HAMK) · Finland

Improved educational outcomes for 50+ students and developed curriculum. Provided personalized guidance and technical mentoring.

Python · Teaching · Curriculum Development

Lead Machine Learning Engineer

2024

Omdena Kitwe, Zambia · Remote

Delivered a COVID-19 forecasting system for the Kitwe, Zambia (Omdena) chapter: XGBoost models, Streamlit app, and standard regression metrics (MAE, RMSE, R²) as in the public repository. Worked with a global team of 15+ contributors.

Team Leadership · XGBoost · Healthcare AI

Machine Learning Engineer

2023 to 2024

Raqameyyah Digital Consultancy · Egypt (Remote)

Built and deployed ML models using advanced techniques. Developed chatbots with advanced prompt engineering and created front-end applications to enhance customer experience.

Machine Learning · Feature Engineering · Application Deployment · Python

Data Analyst

2023

Raqameyyah Digital Consultancy · Egypt (Remote)

Cleaned and preprocessed large datasets for analysis. Developed applications for data filtering and visualization and created automated data processing pipelines.

Data Analysis · Dashboards · Streamlit · EDA

Accounting Intern

2022

foodora Finland · Tampere

Gained experience in financial accounting and supply operations. Worked with Microsoft Excel for data management and contributed to financial reporting processes.

Supply Operations · Financial Accounting · Microsoft Excel

Social Media Representative

2022

Foody Allen · Tampere

Lead content creator for food producer Njunno during Block-festival weekend. Concentrated mainly on creating photo- and videography content from the event for marketing and after-marketing purposes.

Multi-Channel Marketing · Graphic Design · Event Photography · Video Editing

Marketing Communications and Content Production

2022

BSB Fashion · Paphos, Cyprus

Developed marketing strategies and content production. Managed e-commerce platforms and multi-channel marketing and created graphic designs and video content.

Sales · E-Commerce · Multi-Channel Marketing · Graphic Design

Sales Associate

2021 to 2022

Keinuipuiston kukka Oy · Tampere, Finland

Main responsibilities were customer service and sales. Managed the store's online shop and delivering platforms Foodora and Wolt and developed business-to-business relationships.

Business-to-Business (B2B) · Sales · E-Commerce · Multi-channel Retail

Founder

2007 to 2014

Plastic Recycling for Sustainable Solutions · Hybrid

Founded and managed a plastic recycling factory focused on environmental protection. Developed business strategies and operational methods and managed employees and production processes. Promoted environmental protection by transforming plastic waste into valuable resources.

Business-to-Business (B2B) · New Business Development · Supply Operations · Environmental Sustainability

Education

Bachelor of Business Information Technology (Data Science and Machine learning) — Grade 4.7

Häme University of Applied Sciences (HAMK), Finland · 2023 to 2026

Specialization in Data Science and Business Intelligence. Focus on machine learning, data analytics, and business applications.

Thesis (BITCA): Microsoft-based performance goal management system for strategy implementation. PDF on the portfolio site. Catalogue: HAMK Finna (vanaicat.144542).

Climate Change AI (CCAI) Summer School

Climate Change AI (CCAI) · 2024

Intensive program on machine learning and AI for climate change mitigation, adaptation, and environmental decision-making. Certificate: Climate_Change_AI_Sawsan_Abdulbari.pdf on the portfolio site.

Google Data Analytics Specialization

Google · Nov 2023

Courses: Foundations: Data, Data, Everywhere; Ask Questions to Make Data-Driven Decisions; Prepare Data for Exploration; Process Data from Dirty to Clean; Analyze Data to Answer Questions; Share Data Through the Art of

Visualization; Google Data Analytics Capstone: Complete a Case Study. Certificate PDF: Google_Data_Analytics.pdf on the portfolio site.

Vocational Qualification in Business (Business Administration)

Tampere Vocational College (TREDU), Finland · 2020 to 2022

Business administration and management fundamentals. Project management and organizational skills.

Languages

Arabic · Native **English** · Professional proficiency **Finnish** · Professional proficiency

Project references

<https://sawsanabdulbari.github.io/portfolio/#projects> — full project collection with live demos

Skills

Machine Learning

Predictive Modeling · Time Series Analysis · RAG Architecture · Model Deployment · Prompt Engineering

Leadership

Team collaboration (15+ people) · Cross-cultural Collaboration · Mentoring & Training · Project Management · Stakeholder Communication

Data Science

Data Mining & Analytics · EDA & Visualization · Feature Engineering · Statistical Analysis · Dashboard Creation

Business

Strategic Planning · Data Analysis · Process Optimization · Financial Management · Business Development · Operations Management

Selected publications

- **Microsoft-based Performance Goal Management System for Strategy Implementation** — Bachelor's thesis, Häme University of Applied Sciences (2026). Catalogue: HAMK Finna (vanaicat.144542). PDF on the portfolio site.

Personal interests

- **Hiking** — Nature exploration, outdoor activities
- **Fitness** — Health and wellness, physical activity
- **Food Exploration** — Culinary experiences, cultural cuisine
- **Continuous Learning** — Technology trends, professional development
- **Environmental Protection** — Green initiatives, environmentally friendly practices

Portfolio CV, compiled from the live site at sawsanabdulbari.github.io/portfolio. Links and claims are self-declared.